

AI media authentication

The paper proposes a hybrid, multi-layered framework for authenticating digital media and detecting AI-generated content (deepfakes). relying on any single detection method is structurally flawed, as adversaries can easily exploit individual vulnerabilities.

the framework integrates three primary pillars:

1. Cryptographic Content Provenance (C2PA)

C2PA (Coalition for Content Provenance and Authenticity) is an open technical standard used to record the **origin, creation process, and editing history of digital content** such as images, videos, audio, and documents.

How C2PA works

When content is created or edited, C2PA can add a **tamper-evident Content Credential** containing information such as:

- **Who/what created it** — camera, software, AI model, etc.
- **When it was created**
- **What tools were used**
- **What edits were performed**
- **Whether AI was involved**
- **Previous versions or actions performed on the content**

2. Invisible Digital Watermarking:

it acts as the **second layer of authentication**, especially when C2PA metadata is removed.

Invisible digital watermarking means embedding a hidden piece of information directly into the **pixels of an image, frames of a video, or waveform of audio** so that it is not perceptible to humans but can later be detected by a verification system.

What exactly is stored in the watermark?

The watermark can contain information that helps identify the content or its provenance.

For example:

Content ID: 8F72A91

Creator: AI Generator X

Model: Model-X

Timestamp: 2026-08-28

Provenance reference: ABC123

3. Deep Learning Forensics

third layer of defense in your framework

Its job is to examine the **actual content itself**—the pixels, frames, motion, textures, etc.—and look for subtle patterns that suggest the media was **AI-generated or manipulated**.

C2PA looks at the content's history, watermarking looks for a hidden identity signal, while deep-learning forensics looks at the content itself for evidence of manipulation.

How does it detect AI-generated content?

AI generators don't always produce perfectly natural content.

They can leave subtle **generative artifacts** that humans may not notice.

For an image, these might involve:

- unusual texture patterns
- inconsistent fine details
- unnatural edges
- strange frequency patterns
- inconsistent lighting
- unusual facial details
- artifacts around objects
- statistical differences from natural images

XceptionNet is a CNN architecture that can be used for image classification and has been widely explored in deepfake detection.

It is particularly useful because it can learn detailed spatial patterns from images.